

Physics I

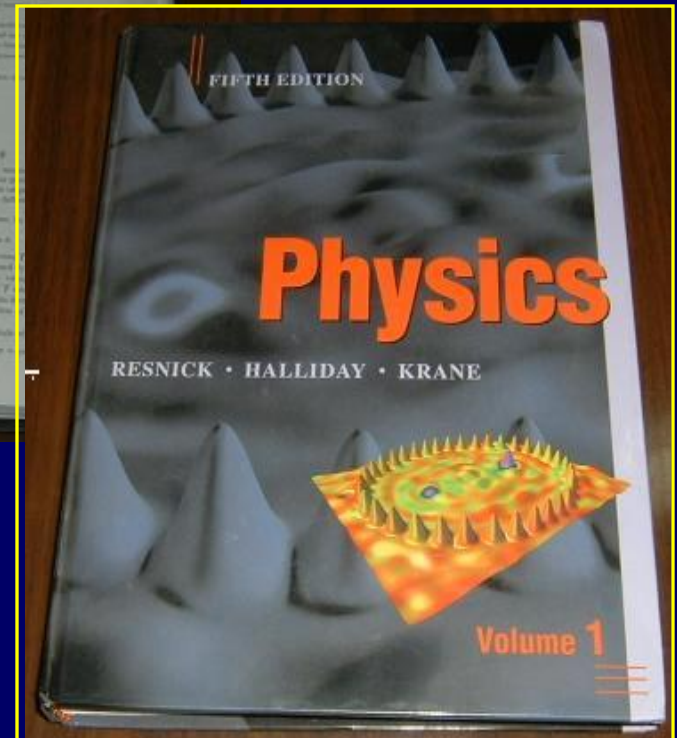
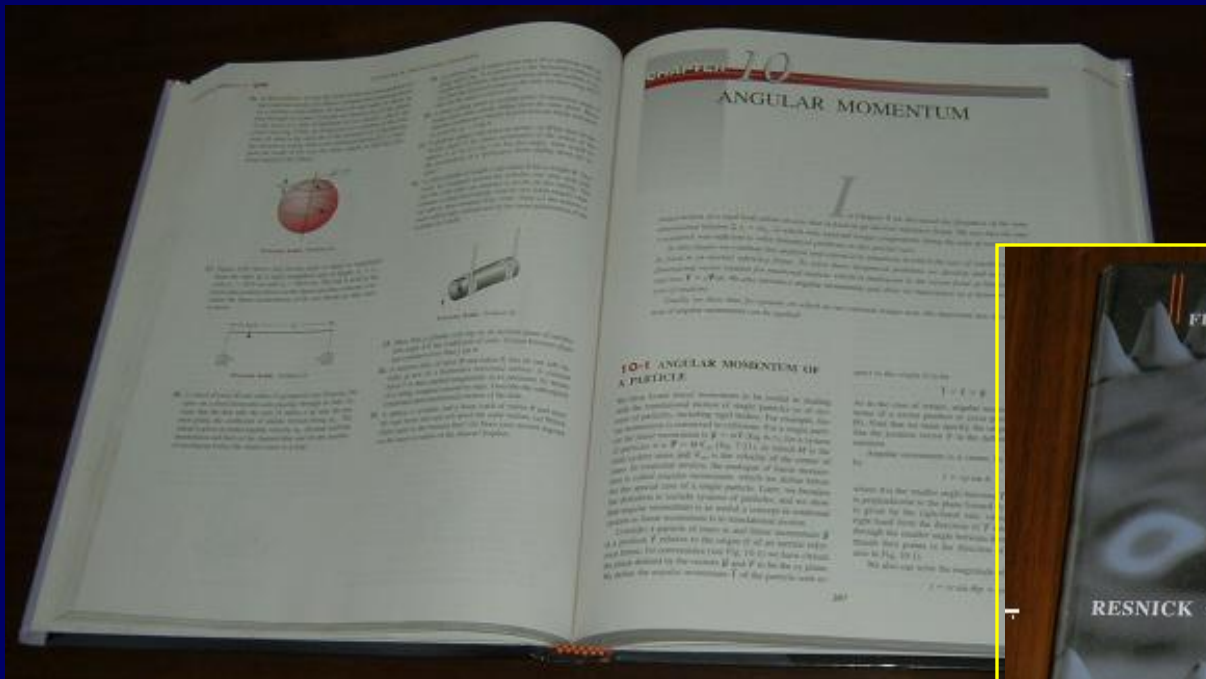
Prof. Zhiwei Ma

马志为

浙江大学聚变理论与模拟中心

Institute for Fusion Theory and
Simulation

Email: zwma@zju.edu.cn



PHYSICS

Resnick,
Halliday,
Krane

Welcome to

Physics I



There are several components:

- **Rehearse** (reading assignments) due before lecture
- **Lectures** (presentations, demonstrations)
- **Quiz** (10%+5% in total)
- **Homework** (15% in total)
- **Mid Exam** (20~30% in total)
- **Final Exam** (50~40% in total)
- **Labs** (hands-on interactions with the phenomena)
- Do not miss lab;

The Challenges of Physics I

1. The course is a few of NEW concepts!!

- Many concepts have been learnt during a high school.
- These old concepts are extended.

2. Math is the language of physics and here you need to learn to work with it

- calculus: line integrals, surface integrals, gradients
- vectors: addition, dot-product, cross-product, decomposition

How to Succeed in Phys I.....

1. Rehearse, and exercises *before* lecture
 - I'll keep them to a minimum
2. Don't get behind!!
3. Ask questions *early* if you don't understand things
 - ask me in lecture !! ...also see me after class.
 - office hours in my office (紫金港海纳苑8幢物理楼419)
 - TA's Office Hours: Sunday 2-5PM in every two weeks (Teacher's lounge Nearest 紫金港东1B-204)
 - Through DingDing.
4. Use problem solving to develop conceptual understanding
 - this is how you really learn science...
5. *Your attitude, not your aptitude, will determine your altitude.*

World Year of Physics 2005

Einstein in the 21st Century



Help make 2005 another *Miraculous Year!*

Timed to coincide with the 2005 Centennial Celebration of Albert Einstein's Miraculous Year, the World Year of Physics 2005 will bring the excitement of physics to the public and inspire a new generation of scientists.

2005世界物理年

物理学是研究物质性质、运动规律及其相互作用的学科，是一项激动人心的智力探险活动，并为人类文明作出巨大贡献。

物理学拓展我们认识自然的疆界，扩展和提高我们对其他学科的理解，是技术进步最重要的基础。

物理教育为科学和技术培养训练有素的人才。

物理学的进步对社会发展和人类生活的改善有不可估量的影响。

主办单位：中国物理学会

支持部门：中国工程院、科技部、教育部、

中国科学院、中国工程院、国家自然科学基金委员会



推动人类文明的物理学

——世界物理年2005

前言

1905年，爱因斯坦创建了狭义相对论，改变了时间和空间的观念。也是在1905年，爱因斯坦首先用量子论解释光电效应，揭示了光的波动性和粒子性的统一。这一年，他还发表了另外几篇重要的学术论文。1905年成为爱因斯坦的“奇迹年”。因此，国际纯粹与应用物理联合会（IUPAP）从2003年就开始筹划在2005年举办全球性的“世界物理年”（World Year of Physics 2005）庆祝活动，以提高公众对物理学的了解。

相对论和量子论从根本上改变了人类的宇宙观，也为20世纪物理学的发展奠定了基础。回顾上个世纪，物理学在半导体、集成电路、激光、超导等方面的发现是信息革命的科学基础，由物理学研究衍生的新技术和新产品层出不穷，深刻地改变了人们的生产方式和生活方式。它还为了生物、医疗、地学、农业提供了强大的探测手段和研究方法，促进了化学、天文、材料、能源、信息等学科的发展。

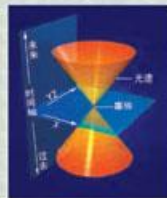
联合国大会于2004年6月10日通过决议，正式宣布2005年为“国际物理年”（International Year of Physics）。这理所当然地引起了世界各国人民更加认真看待物理学在自己的日常生活和探索自然与宇宙的基本问题中的重要性。

中国物理学会在中央有关部门的大力支持下，积极组织并参与这一全球性纪念活动。为此，我们编写了这套图文资料，供各地在举办相关活动时使用。

世界物理年徽标：其构图像一个光锥（注1）。它的红底代表过去。在物理学中，光谱发生“红移”（注2）是物体向远离我们的方向运动的光谱特征；红色又有底部或基础的含义。它的蓝顶表示未来，天空是蓝色的；“蓝移”是面向我们方向运动物体的光谱特征。黄和绿连接着过去与未来，体现出在过去的基础上建立未来的信心。其中，绿色又代表“绿灯可走”，即进步；黄色代表和平、合作。因此，这个徽标主要表达的意义是：技术进步和国际合作共建光明的未来。只要我们大家，特别是物理工作者向这个目标努力，就能够得到有益于社会的新发现。

[注1]光锥：将一块石头扔到池塘里，水面的涟漪以圆周的形式散开并越变越大。如果使用“空间-时间图”，向上增加的方向是时间，水平方向是池塘水面，这些扩大的水波的圆圈就画出一个圆锥，其顶点即为石头击到水面的地方和时间。类似地，从发生于特定时刻和空间中特定一点的一个事件散开的光在四维的空间——时间里形成了一个三维的圆锥，这个圆锥称为事件的过去光锥。我们在天空中所见到的任何一个天体都是它在发光瞬间的像。天体离得越远，我们今天见到的像就在时间上就退得越早。同样道理，一个事件将产生一个未来光锥，事件以光速向我们逼近。光锥可以确保起因在前，而结果在后。

[注2]红移和蓝移：当光源远离观测者运动时，接收到的光波频率比其固有频率低，即向红端偏移，这种现象称为“红移”。当光源向着观测者运动时，接收频率增高，相当于向蓝端偏移，称为“蓝移”。哈勃发现，来自星系的光谱呈现某种系统性的红移，即星系正在远离我们。



主办单位：中国物理学会

支持部门：中国工程院、科技部、教育部、中国科学院、中国工程院、国家自然科学基金委员会

zwma@zju.edu.cn

www.cps.org.cn cps@aphy.iphy.ac.cn



推动人类文明的物理学

—— 世界物理年2005

理论篇——永恒的追求

牛顿力学：

牛顿之前的亚里士多德理论认为支配天体运行的规律（天界）和支配地面物体运动的规律（世俗）是完全不同的。牛顿运动定律和万有引力定律统一了天体和地面物体的运动。天上的月亮绕地球运动和地上苹果落地都是由地球的引力引起的，遵从同样的规律。它是物理学第一次大统一。

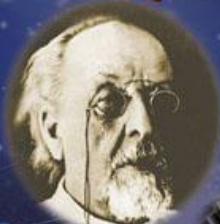
牛顿第二定律 $f=ma$
万有引力定律 $f=km_1m_2/r^2$



不朽科学名著——
《自然哲学的数学原理》



宇宙速度



齐奥尔科夫斯基



神州五号与杨利伟



登月

1846年，科学家根据天王星运动的异常用牛顿力学算出在其轨道之外还有一颗大行星，天文学家随即在他们指出的天区发现了海王星。这是牛顿力学的辉煌成就。物理学的一个基本信念是：自然界是统一的，物理世界存在着完整的因果链条。牛顿力学是体现这一信念的第一个成功的范例，因而成为物理学发展的起点。



推动人类文明的物理学

—— 世界物理年2005

应用篇——无所不在 无所不用

激光器的诞生



汤斯



肖洛



梅曼及第一台激光器

激光是物质受激辐射发出的光，激光有许多奇异特性：相干性好，单色性好，方向性好，亮度高等。激光技术的原理基于爱因斯坦1916年提出的受激辐射概念。1958年，美国科学家汤斯和肖洛给出了制造激光器的理论分析和设计方案。1960年，美国科学家梅曼制造出第一台激光器——红宝石激光器。40多年来，激光技术和激光应用均以极快的速度发展。

主办单位：中国物理学会 中国物理学会

www.cps.org.cn cps@aphy.iphy.ac.cn

支持部门：2026-3-20，科技部、教育部、中国科学院、中国工程院、国家自然科学基金委员会

主办单位：中国物理学会 中国物理学会

www.cps.org.cn cps@aphy.iphy.ac.cn

中国科学技术协会、科技部、教育部、中国科学院、中国工程院、国家自然科学基金委员会

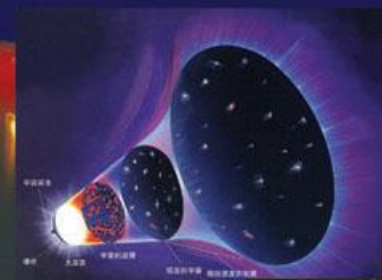
Physics 2005

推动人类文明的物理学

—— 世界物理年2005

展望篇——热点和展望

宇宙尺度上，宇宙演化的规律，宇宙标准模型理论的检验，暗物质和暗能量寻求形成当前热点。整个宇宙至少90%是暗物质，而不是已知的物质。探测暗物质的性质有助于探索自然的基本和宇宙的结构和演化之间的关系。



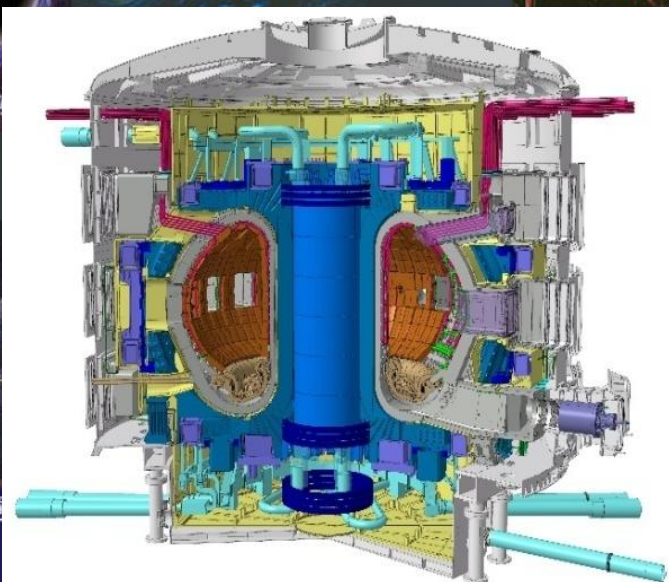
宇宙演化



暗



哈伯望远镜



磁约束聚变反应堆ITER



百家号/萌呆冰



爱因斯坦 (A. Einstein, 1879—1955)
相对论的创建者，量子论的奠基人之一

生于德国乌尔姆市，卒于美国普林斯顿。1905年同时在狭义相对论、量子理论和布朗运动三方面作出重大成就，揭示了时间和空间的统一性、力学理论和电磁理论在运动学上的统一性，揭示了质能相当定律，即 $E=mc^2$ ；首先用量子论解释光电效应，揭示了光的波动性和粒子性的统一；解释了布朗运动的热力学本质，以多种方式证实了分子和原子的实在性。因此，1905年被称为人类历史上的“奇迹年”。1915年，创建广义相对论，深刻揭示了时空和物质的统一性，指出空间的几何性质取决于物质分布，它不是平坦的欧几里得空间，而是弯曲的黎曼空间，弱引力场中沿直线而沿曲线传播。由此，引发人类历史上一场科学、哲学和思想方法的伟大变革。此后，专致于建立统一理论。

爱因斯坦一生为反对核战争努力，呼吁世界和平作出不懈努力。1922年曾访问上海，对当时中国人的悲惨境况表示极大同情，对西方列强欺凌中国人表示愤慨。抗日战争爆发后，呼吁世界制止战争，声援中国，并在美国上千个城市发起给华捐款运动。

*For k_2 in the neighborhood $S = 17 - k_2$.
The elegant $U_2 \rightarrow k_2$ is 11 to the Temperature k_2 and k_2 can be easily constructed. This gives the solution.*

办



主办单位:  中国物理学会 www.cps.org.cn cps@aphy.iphy.ac.cn
支持部门: 中国科学技术协会、科技部、教育部、中国科学院、中国工程院、国家自然科学基金委员会

主办单位:  zwma@zju.edu.cn
支持部门: 中国科学技术协会、科技部、教育部、中国科学院、中国工程院、国家自然科学基金委员会

2026-3-20

☞ Curiosity to natural phenomena

☞ Thinking, Questions, Trying to find answer



Where do the wind come?

Why the snow is white?

No thinking → No question → No progress!



PHYSICS

- Mechanics
- Fluid Mechanics
- Wave Dynamics
- Thermodynamics
- Special Theory of Relativity
- Electromagnetism
- Optics

PHYSICS

- **Mechanics**
- **Fluid Mechanics**
- **Wave Dynamics**
- **Thermodynamics**
- **Special Theory of Relativity**

PHYSICS 1

T, p, V



$\bar{v}, \bar{K}, \bar{E}$

average

distribution

- **Mechanics**
- **Thermodynamics**

$f(v), f(E)$

Force law

environment

force

Many particles In random

body

motion

Motion law

Experiment is the base of PHYSICS

Chap.1 Measurement

Experiment is the base of PHYSICS

Physical Quantities

Length, mass, time, force,
Speed, density, temperature
Current, magnetic field strength

Experiment is the base of PHYSICS

Physical Quantities

Length, mass, time, force,
Speed, density, temperature
Current, magnetic field strength

1-2 The International System of Units — SI

TABLE 1-1 SI Base Units

<i>Quantity</i>	<i>SI Unit</i>	
	<i>Name</i>	<i>Symbol</i>
Time	second	s
Length	meter	m
Mass	kilogram	kg
Amount of substance	mole	mol
Thermodynamic temperature	kelvin	K
Electric current	ampere	A
Luminous intensity	candela	cd

$$1 \text{ N} = 1 \text{ kg m/s}^2$$

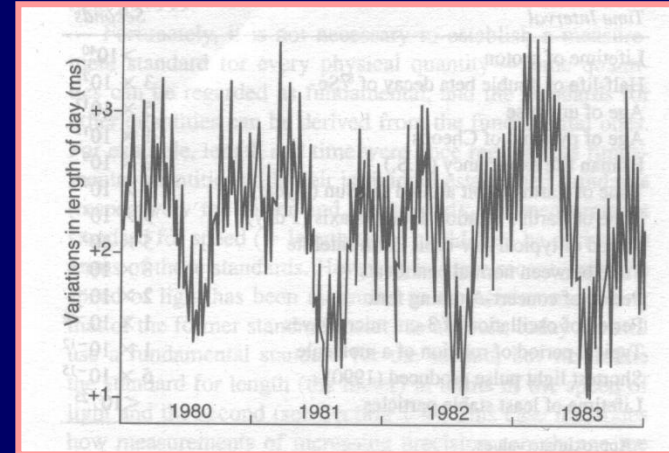
TABLE 1-2 SI Prefixes^a

<i>Factor</i>	<i>Prefix</i>	<i>Symbol</i>	<i>Factor</i>	<i>Prefix</i>	<i>Symbol</i>
10^{24}	yotta-	Y	10^{-1}	deci-	d
10^{21}	zetta-	Z	10^{-2}	centi-	c
10^{18}	exa-	E	10^{-3}	milli-	m
10^{15}	peta-	P	10^{-6}	micro-	μ
10^{12}	tera-	T	10^{-9}	nano-	n
10^9	giga-	G	10^{-12}	pico-	p
10^6	mega-	M	10^{-15}	femto-	f
10^3	kilo-	k	10^{-18}	atto-	a
10^2	hecto-	h	10^{-21}	zepto-	z
10^1	deka-	da	10^{-24}	yocto-	y

Standards and Units

Time (second, s)

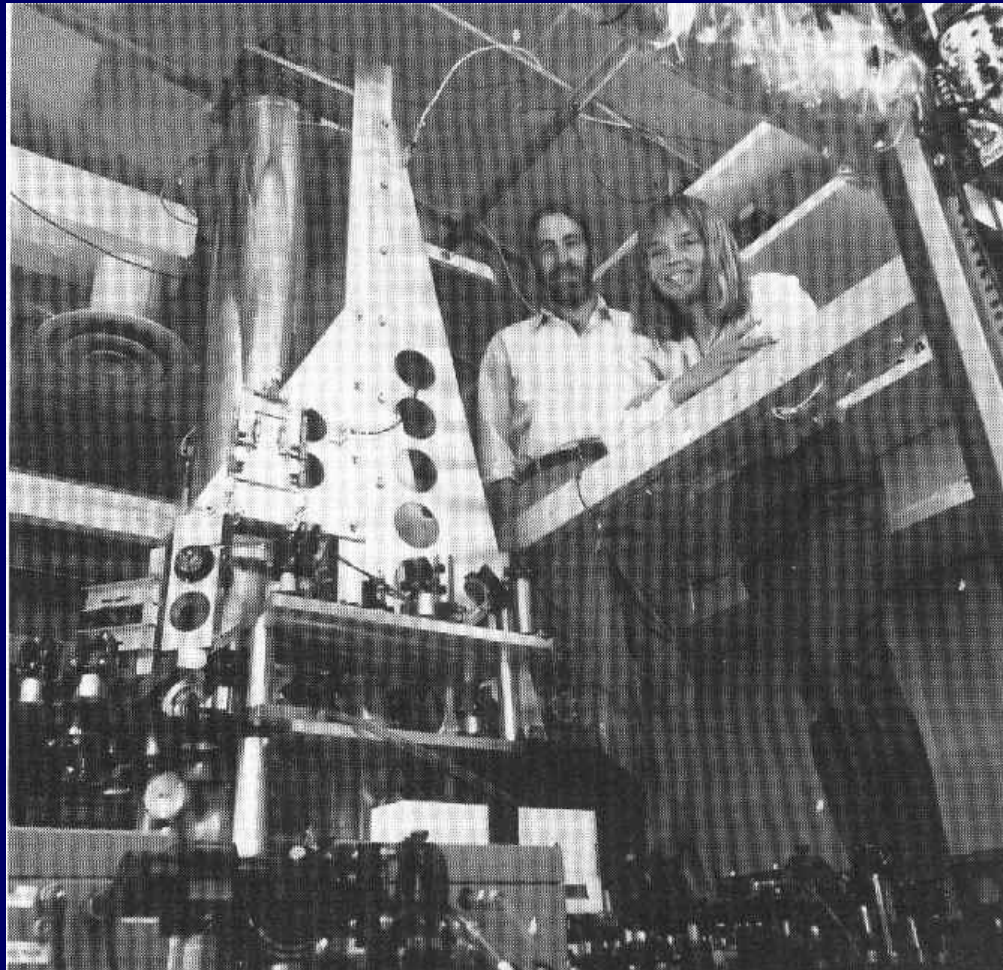
$$\frac{1 \text{ year}}{365 \times 24 \times 60 \times 60} = 1 \text{ s}$$



**1 s = 9192631770 vibration
of radiation emitted by
a cesium atom**

1-3 The Standard of Time

NIST-F1





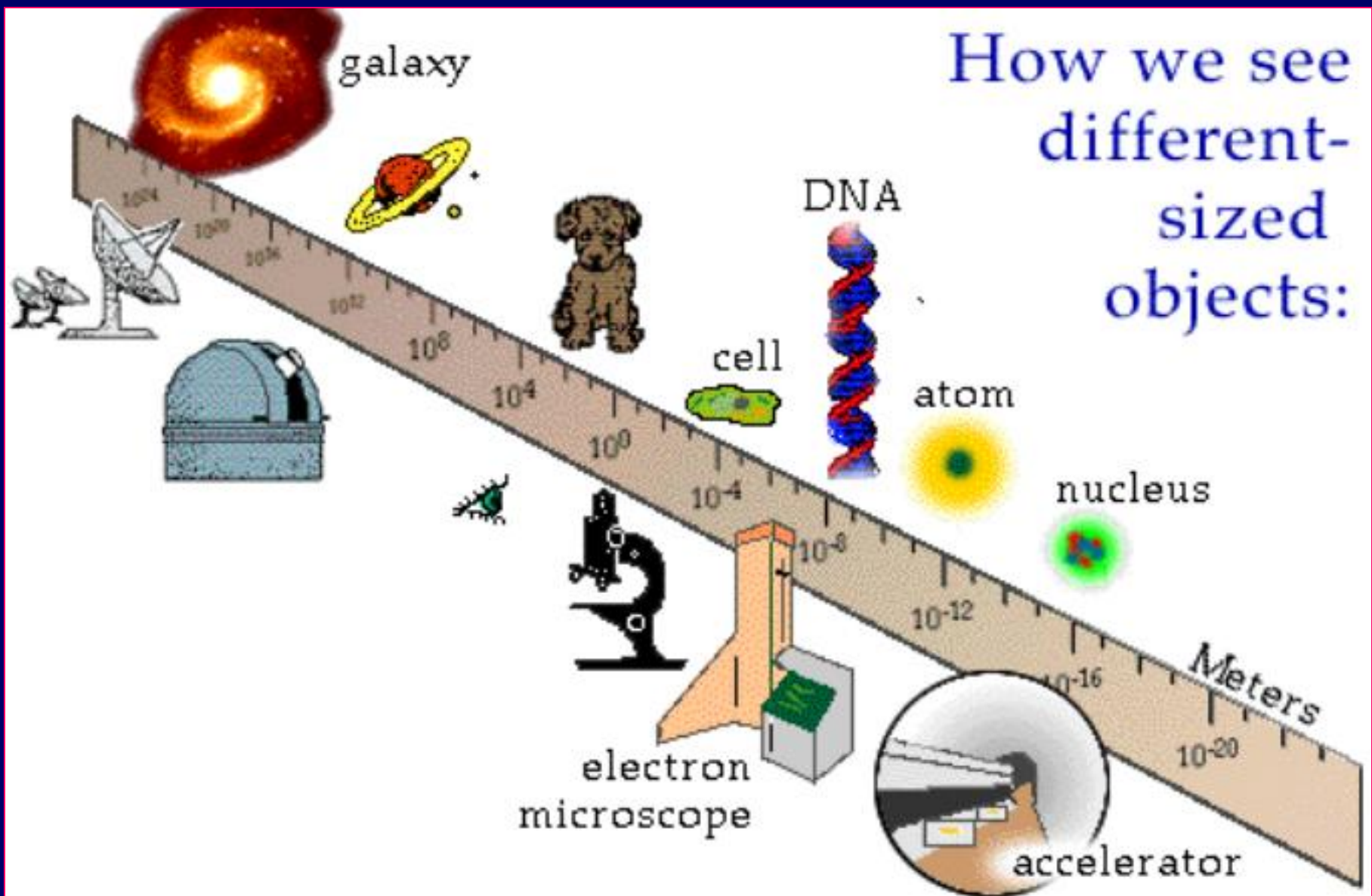
Standards and Units

Length (meter, m)



Traveled by light in vacuum
during a time of $1/299792456$
second

How we see different- sized objects:



10^{-3} 10^{-4} 10^{-5} 10^{-6} 10^{-7} 10^{-8} 10^{-9} 10^{-10} m



Microscope



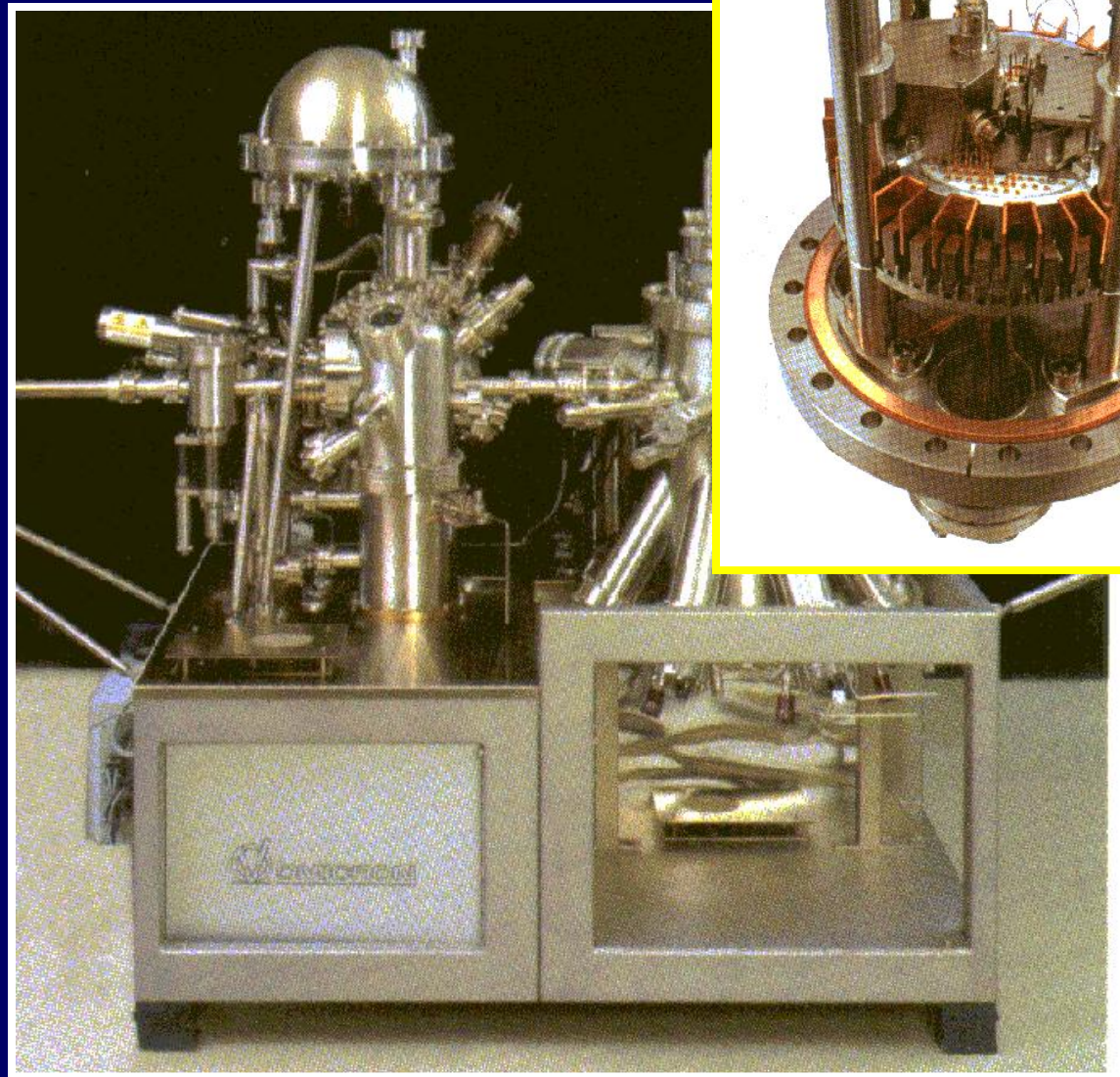
Electron Microscope

Scanning Tunneling Microscopes
(STM)

Size of atom



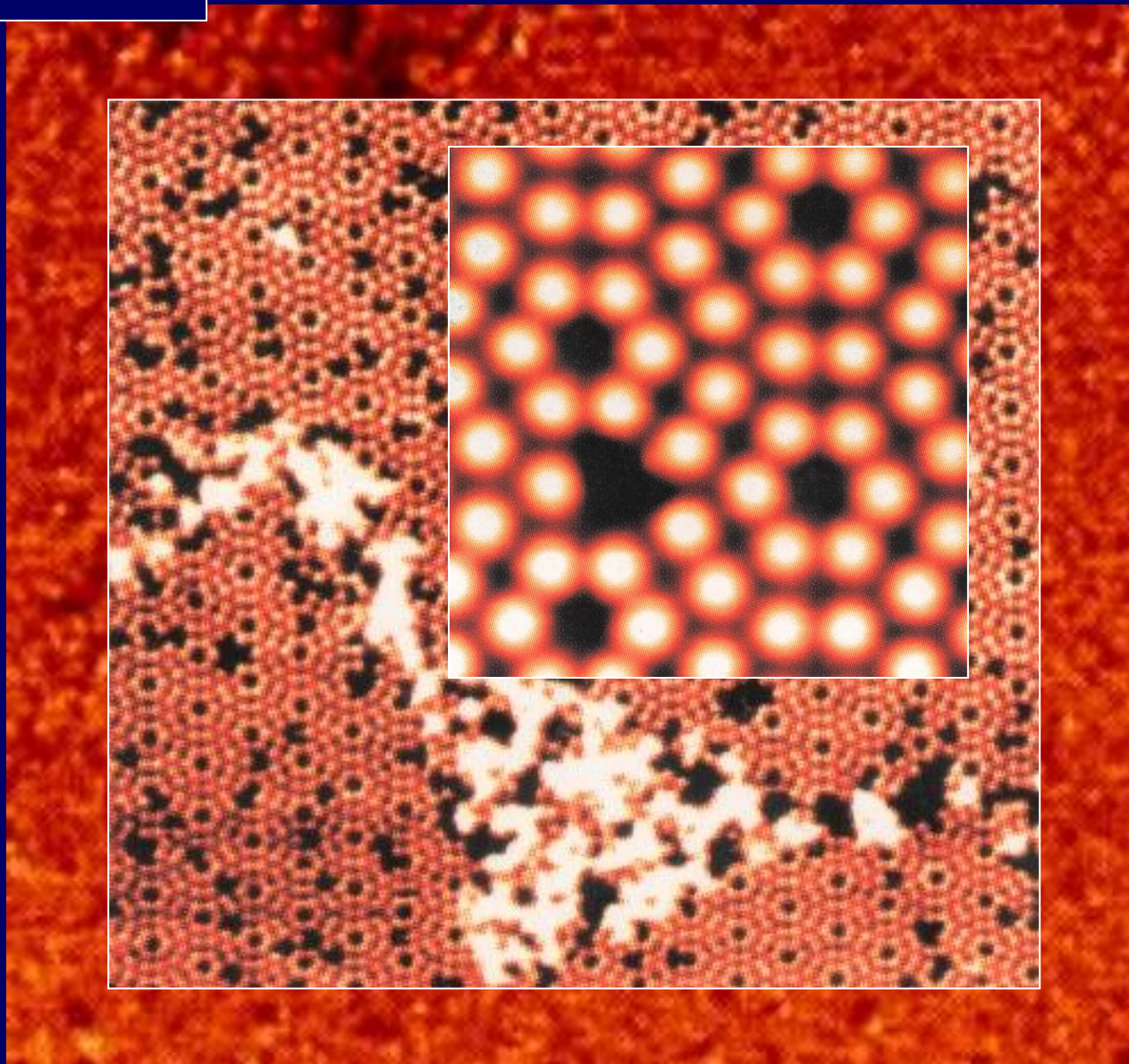
Scanning Tunneling Microscopes (STM)



Si(111) Surface



Si(111) Surface



Non-standards Units

Length

1 yard = 0.9144 meter

1 inch = 2.54 centimeters

1 light-year = 9.48×10^{15} m

Standards and Units

2018-11-16, the 26th General Conference of Weights and Measures passed the resolution of “Amendment of the International System of Units” and officially updated the definition of four basic units including the “Kilogram” of the international standard quality unit. The fully redefined international unit of measurement system will be formally implemented on May 20, 2019, from the World Metrology Day

Mass (kilogram, Kg)

It is defined a fixed value of the Planck constant h to be

$$6.6260701475 \times 10^{-34} \text{ kg} \cdot \text{m}^2 \text{s}^{-1}$$

Galaxy	10^{43} kg
sun	10^{30} kg
Earth	10^{24} kg
person	10^1 kg
virus	10^{-15} kg
atom	10^{-27} kg
electron	10^{-31} kg

Non-standards Units

Mass

1 pound = 0.454 Kg

1 ounce = 0.02835 Kg

1-6 Precision and Significant figures

Rule 1: keep all digits up to the first doubtful one

Rule 2: the number of significant figures should not be enlarged in multiplying or dividing

Rule 3: In adding and subtracting, similar to rule 1

Dimension and dimensional analysis

M, L and T

mass, Length and time

$$[x]=L \quad [v]=LT^{-1}$$

$$[a]=LT^{-2} \quad [f]=MLT^{-2}$$

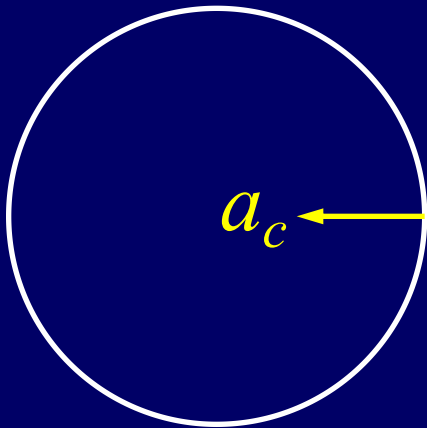
$$[Q] = L^a M^\beta T^\gamma \quad a, \beta, \gamma$$

→ dimensional exponents

Example

Circular motion

Centripetal acceleration $a_c = v^2/r$



$$LT^{-2} = [a_c] = \frac{[v^2]}{[r]} = \frac{(L/T)^2}{L}$$

Example

Free fall

$$LT^{-1} = v \propto g^p h^q = \left(\frac{L}{T^2} \right)^p L^q = L^{(p+q)} T^{-2p}$$

$$1 = p + q \quad -1 = -2p \quad p = \frac{1}{2} \quad q = \frac{1}{2}$$

$$v \propto g^{1/2} h^{1/2} \quad v = \sqrt{2gh}$$

Example

Constant acceleration motion

$$r = r_0 + v_0 t + \frac{1}{2} a t^2$$

$$L \quad L + \left(\frac{L}{T}\right)T + \left(\frac{L}{T^2}\right)T^2 = L + L + L$$

Chapter 2

Motion in 1-D (Kinematics)

Physical Quantities

Length, mass, time, force,
Speed, density, temperature
Current, magnetic field

Vectors and Scalars

Scalars

magnitude

Vectors

both magnitude
and direction

Speed of light in vacuum

$$3.00 \times 10^8 \text{ m / s}$$

Proton mass

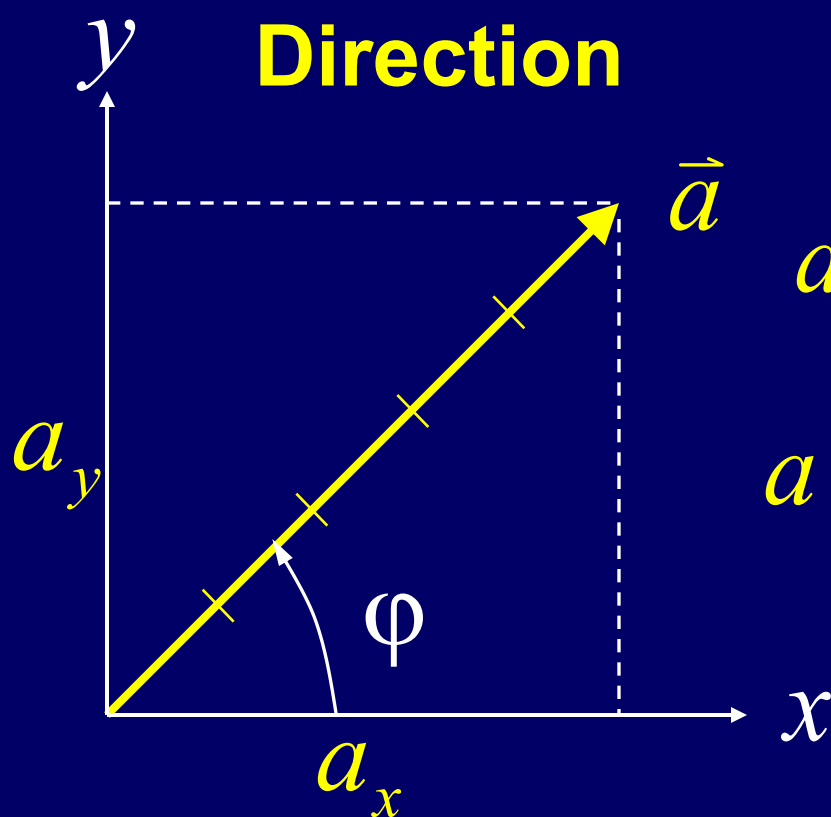
$$1.67 \times 10^{-27} \text{ kg}$$

Vector

magnitude
direction

- To display a Vector
- Vector Calculation
- Vector Application

Properties of vectors



Direction

Magnitude $|\vec{a}|$ or a

$$a_x = a \cos \phi \quad a_y = a \sin \phi$$

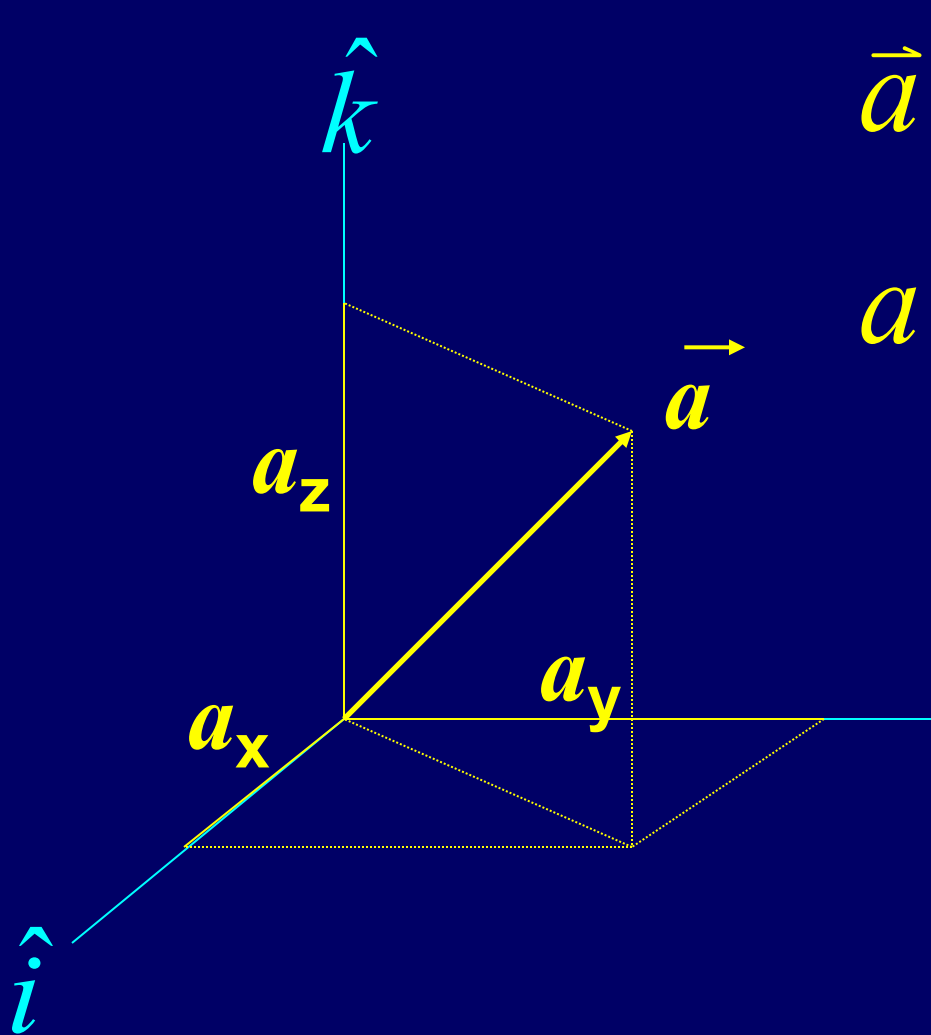
$$a = \sqrt{a_x^2 + a_y^2} \quad \phi = \tan^{-1} \frac{a_y}{a_x}$$

$$\vec{a} = a_x \hat{i} + a_y \hat{j}$$

$$|\hat{i}| = |\hat{j}| = 1$$

Unit vector

Three dimensions



$$\vec{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$$

$$a = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

Magnitude

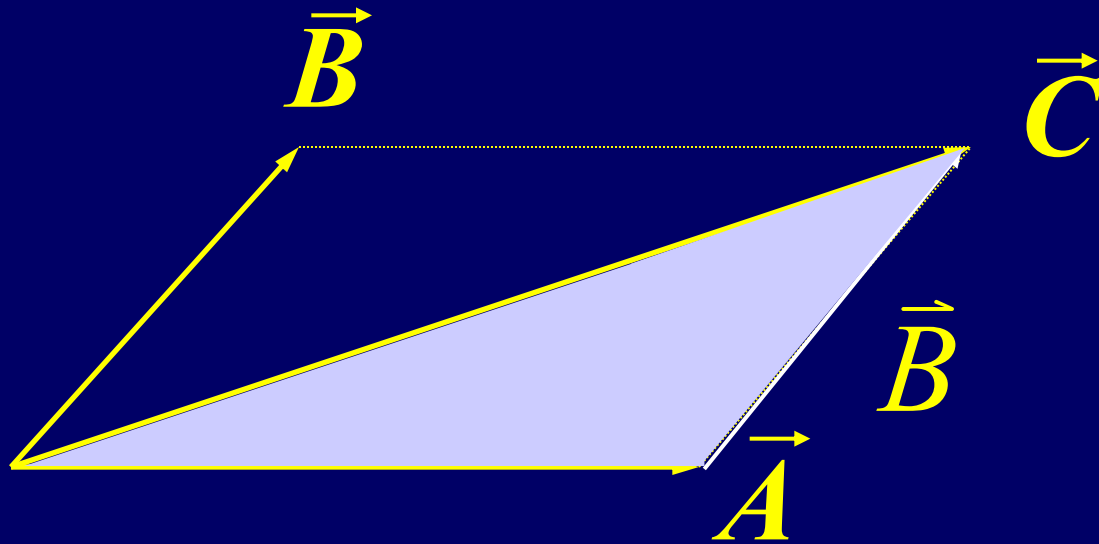
$$\vec{a} = a \hat{u}_a$$

unit vector

$$|\hat{u}_a| = 1 \text{ or } u_a = 1$$

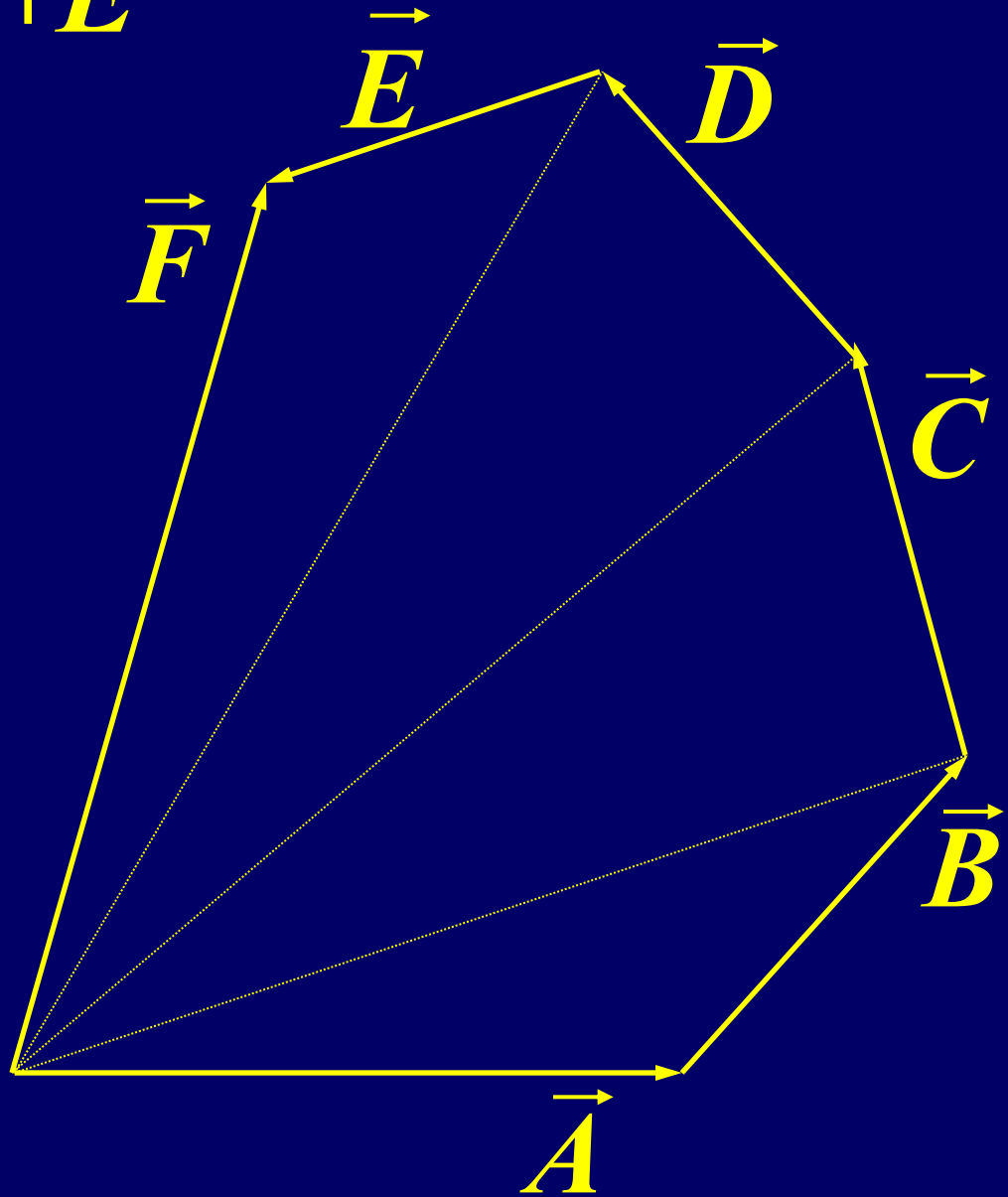
ADDING VECTORS

$$\vec{A} + \vec{B} = \vec{C}$$



Making a Parallel Quadrilateral
Making a Triangle

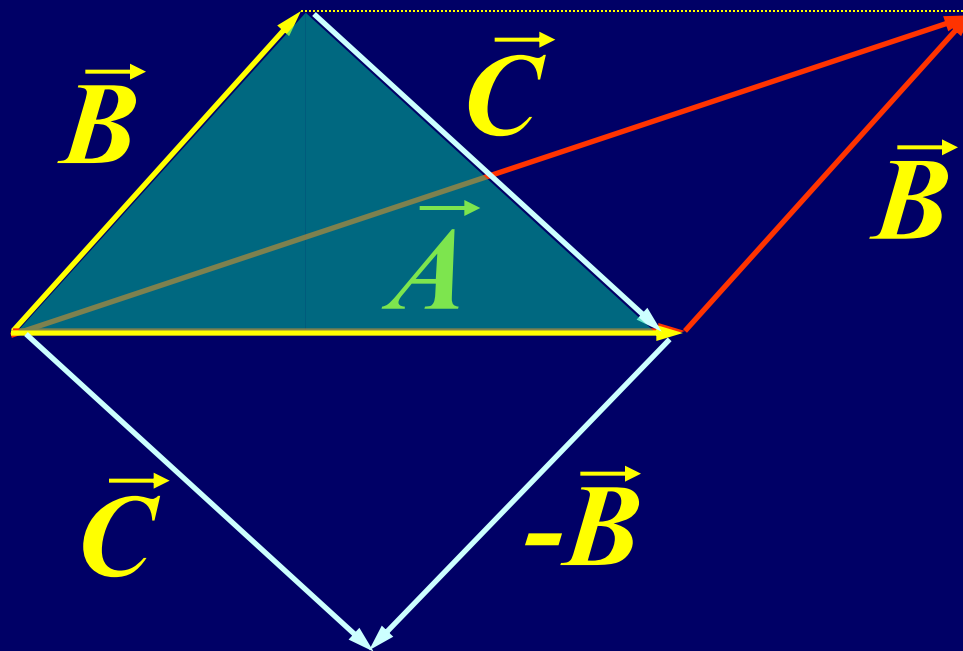
$$\vec{F} = \vec{A} + \vec{B} + \vec{C} + \vec{D} + \vec{E}$$



$$\vec{A}-\vec{B}=\vec{C}$$

$$\vec{A} - \vec{B} = \vec{C} \quad \vec{A} + (-\vec{B}) = \vec{C}$$

$$\vec{A} = \vec{B} + \vec{C}$$



$$\vec{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$$

$$\vec{b} = b_x \hat{i} + b_y \hat{j} + b_z \hat{k}$$

$$\vec{c} = \vec{a} + \vec{b}$$

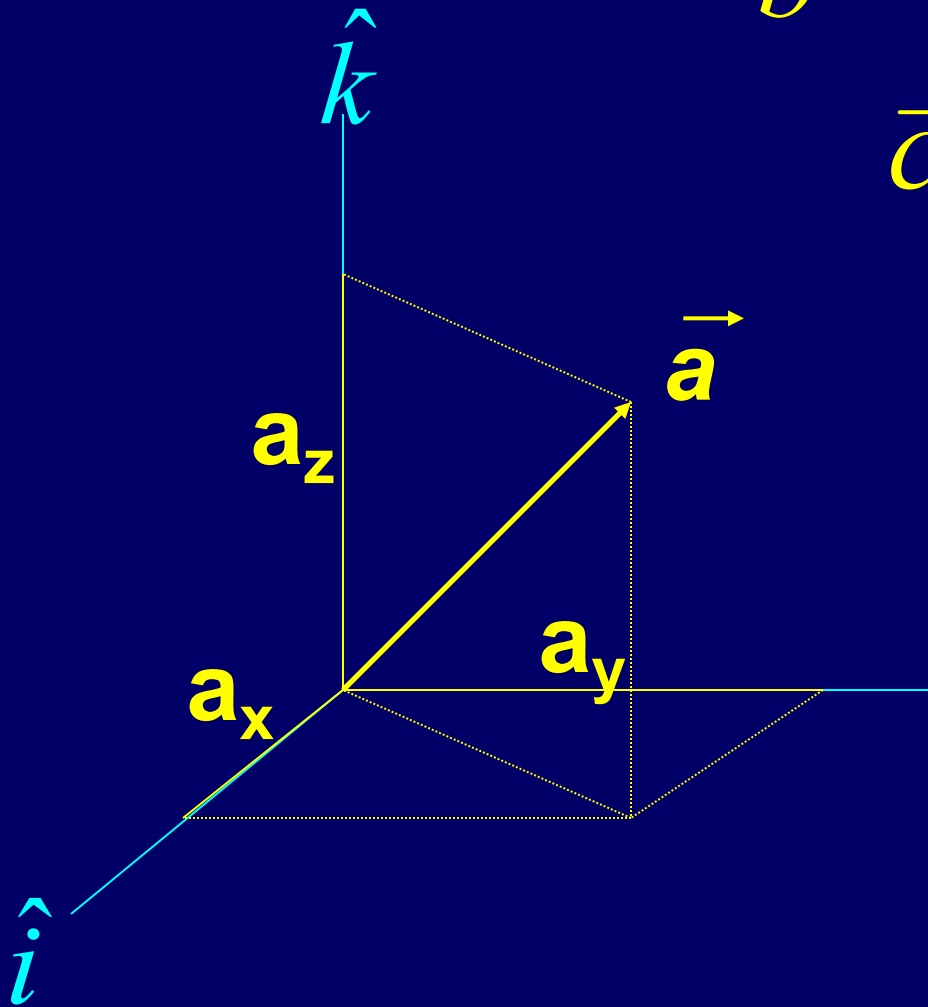
$$= (a_x \hat{i} + a_y \hat{j} + a_z \hat{k})$$

$$+ (b_x \hat{i} + b_y \hat{j} + b_z \hat{k})$$

$$\hat{j} = (a_x + b_x) \hat{i}$$

$$+ (a_y + b_y) \hat{j}$$

$$+ (a_z + b_z) \hat{k}$$



MULTIPLYING OF VECTORS

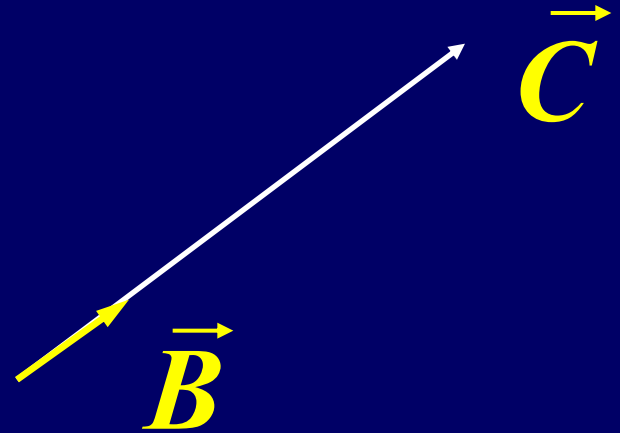
$$a\vec{B} = \vec{C} \quad \text{vector}$$

$$\vec{A} \cdot \vec{B} = c \quad \text{scalar}$$

$$\vec{A} \times \vec{B} = \vec{C} \quad \text{vector}$$

$$a\vec{B} = \vec{C}$$

$$\vec{B} = b_x \hat{i} + b_y \hat{j} + b_z \hat{k}$$



$$\vec{C} = a\vec{B}$$

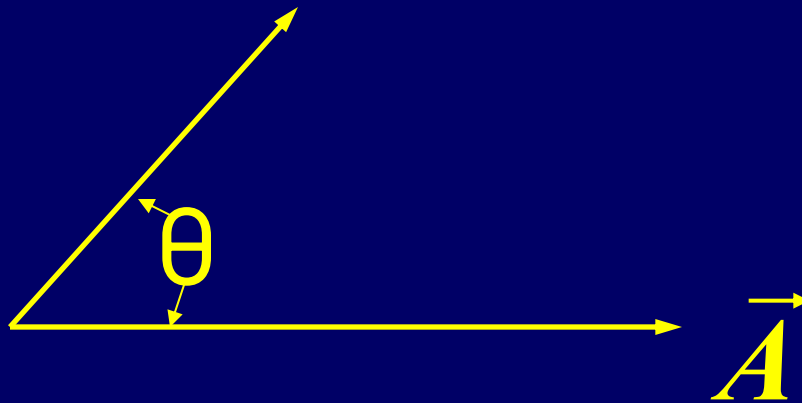
$$= ab_x \hat{i} + ab_y \hat{j} + ab_z \hat{k}$$

Direction no change
Magnitude change

Multiplying of vectors

$$\vec{A} \cdot \vec{B} = c = A B \cos \theta$$

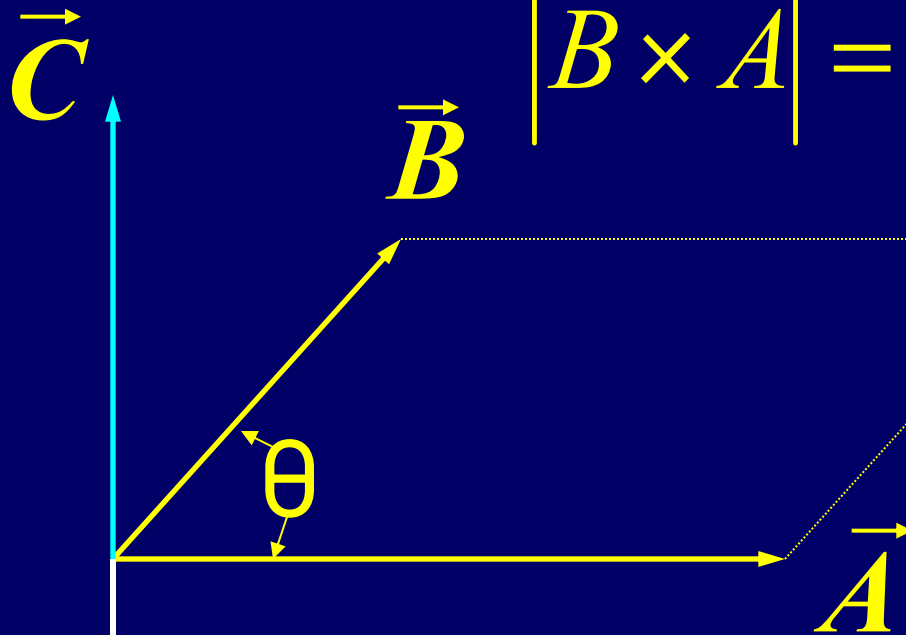
$$\vec{B} \cdot \vec{A} = B A \cos \theta = \vec{A} \cdot \vec{B}$$



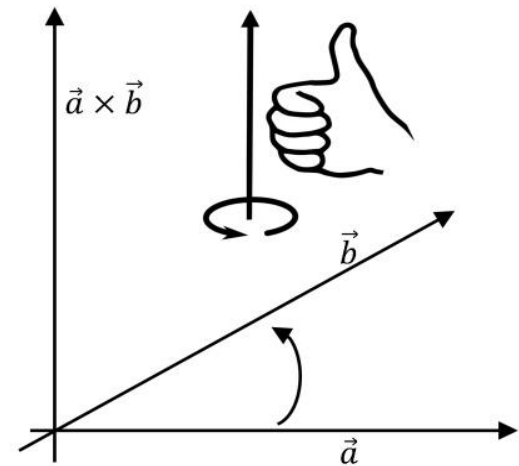
$$\vec{A} \times \vec{B} = \vec{C} \quad |\vec{C}| = A B \sin \theta \quad \vec{B} \times \vec{A} = ?$$

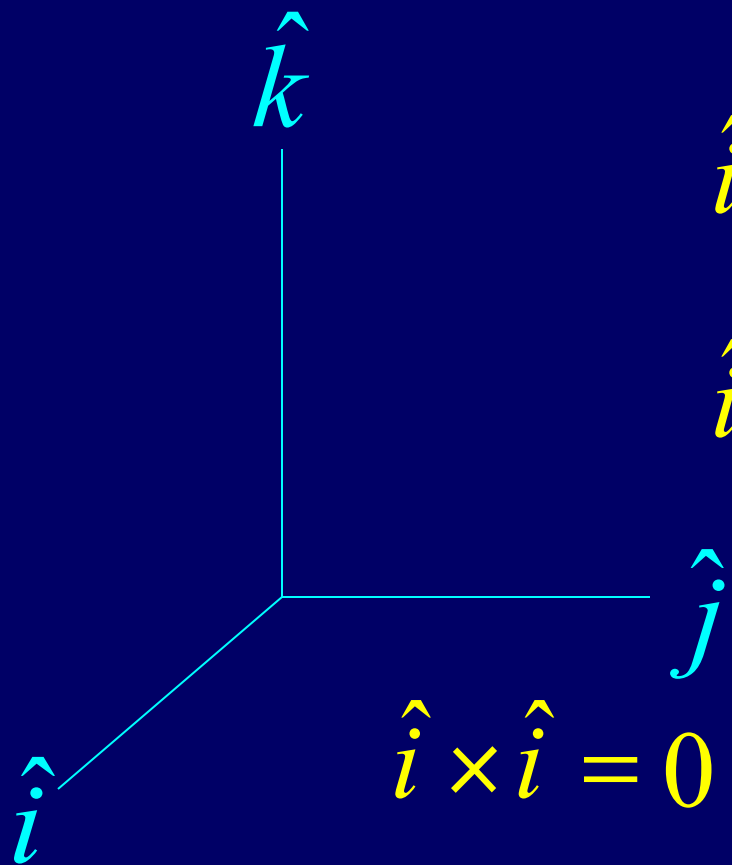
$$|\vec{B} \times \vec{A}| = B A \sin \theta = |\vec{A} \times \vec{B}|$$

$$\vec{B} \times \vec{A} = -\vec{A} \times \vec{B}$$



$$\vec{B} \times \vec{A}$$





$$\hat{i} \cdot \hat{j} = 0 \quad \hat{j} \cdot \hat{k} = 0 \quad \hat{i} \cdot \hat{k} = 0$$

$$\hat{i} \cdot \hat{i} = 1 \quad \hat{j} \cdot \hat{j} = 1 \quad \hat{k} \cdot \hat{k} = 1$$

$$\hat{i} \times \hat{i} = 0 \quad \hat{j} \times \hat{j} = 0 \quad \hat{k} \times \hat{k} = 0$$

$$\hat{i} \times \hat{j} = \hat{k} \quad \hat{j} \times \hat{k} = \hat{i} \quad \hat{i} \times \hat{k} = -\hat{j}$$

Multiplying of vectors

$$\vec{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$$

$$\vec{b} = b_x \hat{i} + b_y \hat{j} + b_z \hat{k}$$

$$c = \vec{a} \cdot \vec{b}$$

$$= (a_x \hat{i} + a_y \hat{j} + a_z \hat{k}) \cdot (b_x \hat{i} + b_y \hat{j} + b_z \hat{k})$$

$$= a_x b_x + a_y b_y + a_z b_z$$

Multiplying of vectors

$$\vec{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$$

$$\vec{b} = b_x \hat{i} + b_y \hat{j} + b_z \hat{k}$$

$$\vec{c} = \vec{a} \times \vec{b}$$

$$= (a_x \hat{i} + a_y \hat{j} + a_z \hat{k}) \times (b_x \hat{i} + b_y \hat{j} + b_z \hat{k})$$

$$= (a_y b_z - a_z b_y) \hat{i} + (a_z b_x - a_x b_z) \hat{j}$$

$$+ (a_x b_y - a_y b_x) \hat{k}$$

$$\vec{A} + \vec{B} = \vec{B} + \vec{A}$$

$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$$

$$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$$

$$(s\vec{A}) \times \vec{B} = \vec{A} \times (s\vec{B}) = s(\vec{A} \times \vec{B})$$

$$\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$$

$$\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$$

$$\vec{A} \cdot (\vec{B} \times \vec{C}) = \vec{B} \cdot (\vec{C} \times \vec{A}) = \vec{C} \cdot (\vec{A} \times \vec{B})$$

$$\vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C})\vec{B} - (\vec{A} \cdot \vec{B})\vec{C}$$

$$\begin{aligned}
\vec{A} \times (\vec{B} \times \vec{C}) &= (a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}) \times [(b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}) \times (c_1 \hat{i} + c_2 \hat{j} + c_3 \hat{k})] \\
&= (a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}) \times [(b_2 c_3 - b_3 c_2) \hat{i} + (b_3 c_1 - b_1 c_3) \hat{j} + (b_1 c_2 - b_2 c_1) \hat{k}] \\
&= (a_2 b_1 c_2 - a_3 b_3 c_1 - a_2 b_2 c_1 + a_3 b_1 c_3) \hat{i} \\
&\quad + (a_3 b_2 c_3 - a_1 b_1 c_2 - a_3 b_3 c_2 + a_1 b_2 c_1) \hat{j} \\
&\quad + (a_1 b_3 c_1 - a_2 b_2 c_3 - a_1 b_1 c_3 + a_2 b_3 c_2) \hat{k}
\end{aligned}$$

$$(\vec{A} \cdot \vec{C}) \vec{B} - (\vec{A} \cdot \vec{B}) \vec{C} \quad \boxed{\vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C}) \vec{B} - (\vec{A} \cdot \vec{B}) \vec{C}}$$

$$\begin{aligned}
&= [(a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}) \cdot (c_1 \hat{i} + c_2 \hat{j} + c_3 \hat{k})] (b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}) \\
&\quad - [(a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}) \cdot (b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k})] (c_1 \hat{i} + c_2 \hat{j} + c_3 \hat{k}) \\
&= (a_1 c_1 + a_2 c_2 + a_3 c_3) (b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}) \\
&\quad - (a_1 b_1 + a_2 b_2 + a_3 b_3) (c_1 \hat{i} + c_2 \hat{j} + c_3 \hat{k})
\end{aligned}$$

In the Cartesian coordinate:

Derivation:

$$\vec{A}(t) = a_x(t)\hat{i} + a_y(t)\hat{j} + a_z(t)\hat{k}$$

$$\frac{d\vec{A}(t)}{dt} = \frac{da_x(t)}{dt}\hat{i} + \frac{da_y(t)}{dt}\hat{j} + \frac{da_z(t)}{dt}\hat{k}$$

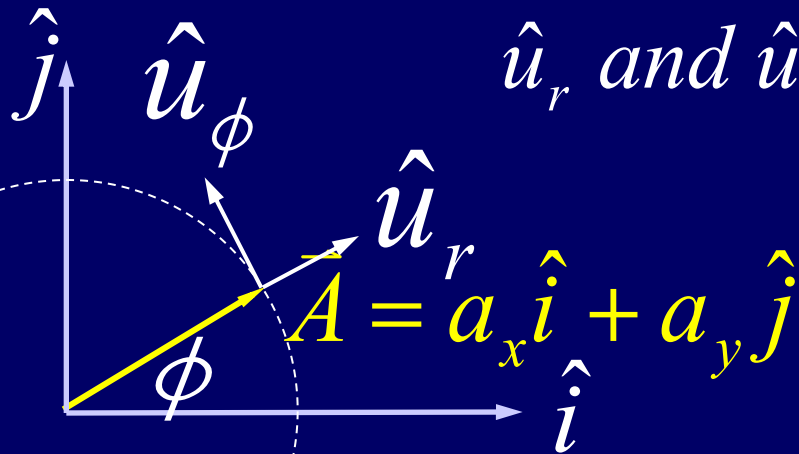
Integral:

$$\int \vec{A}(t)dt = \int a_x(t)dt\hat{i} + \int a_y(t)dt\hat{j} + \int a_z(t)dt\hat{k}$$

plane polar coordinates

$$|\hat{u}_r| = |\hat{u}_\phi| = 1$$

$\hat{u}_r$ and $\hat{u}_\phi$ are the unit vectors



$$\begin{aligned}\hat{u}_r &= \cos \phi \hat{i} + \sin \phi \hat{j} \\ \hat{u}_\phi &= -\sin \phi \hat{i} + \cos \phi \hat{j}\end{aligned}$$

$$\vec{A} = a \hat{u}_r$$

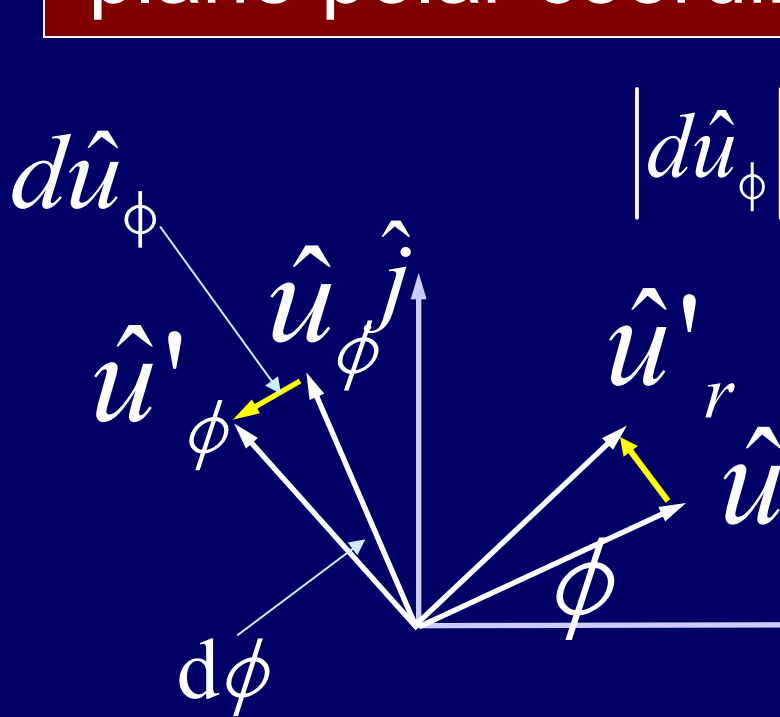
$$\frac{d\vec{A}}{dt} = \frac{da_x}{dt} \hat{i} + \frac{da_y}{dt} \hat{j}$$

$$\frac{d\vec{A}}{dt} = \frac{da}{dt} \hat{u}_r + a \frac{d\hat{u}_r}{dt}$$

$$\frac{d\vec{A}}{dt} = \frac{da}{dt} \hat{u}_r + a \frac{d\phi}{dt} \hat{u}_\phi$$

$$\begin{aligned}\frac{d\hat{u}_r}{dt} &= \frac{d\phi}{dt} (-\sin \phi \hat{i} + \cos \phi \hat{j}) = \frac{d\phi}{dt} \hat{u}_\phi \\ \frac{d\hat{u}_\phi}{dt} &= -\frac{d\phi}{dt} (\cos \phi \hat{i} + \sin \phi \hat{j}) = -\frac{d\phi}{dt} \hat{u}_r\end{aligned}$$

plane polar coordinates



The diagram illustrates the relationship between unit vectors in Cartesian and polar coordinates. It shows a Cartesian coordinate system with unit vectors $\hat{i}$ and $\hat{j}$. A polar coordinate system is defined by unit vectors $\hat{u}_r$ and $\hat{u}_\phi$ at an angle ϕ from the $\hat{i}$ axis. The derivative vectors $\frac{d\hat{u}_r}{dt}$ and $\frac{d\hat{u}_\phi}{dt}$ are shown as small yellow arrows. The angle ϕ is also indicated for the derivative vectors.

$$|d\hat{u}_\phi| = u_\phi d\phi = d\phi \quad \frac{|d\hat{u}_\phi|}{dt} = \frac{d\phi}{dt}$$

$$\frac{d\hat{u}_r}{dt} = \frac{d\phi}{dt} \hat{u}_\phi \quad \frac{d\hat{u}_\phi}{dt} = -\frac{d\phi}{dt} \hat{u}_r$$

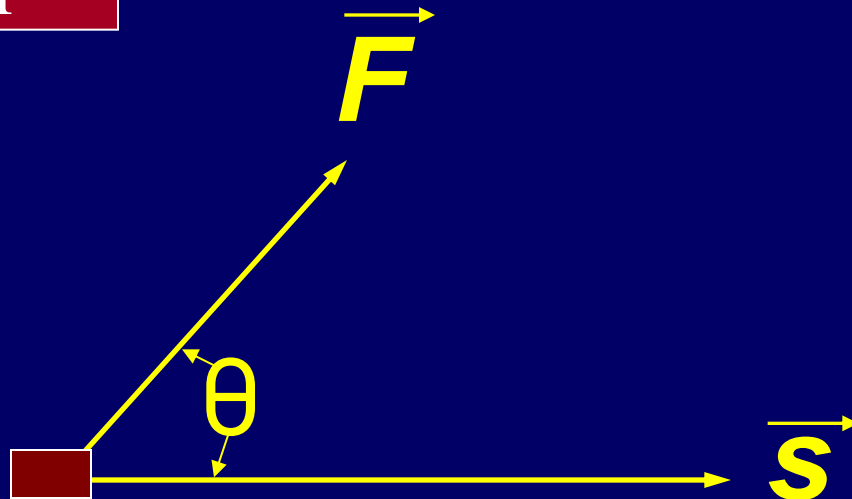
$$\begin{aligned} \frac{d\hat{u}_r}{dt} &= \frac{d\phi}{dt} (-\sin \phi \hat{i} + \cos \phi \hat{j}) = \frac{d\phi}{dt} \hat{u}_\phi \\ \frac{d\hat{u}_\phi}{dt} &= -\frac{d\phi}{dt} (\cos \phi \hat{i} + \sin \phi \hat{j}) = -\frac{d\phi}{dt} \hat{u}_r \end{aligned}$$

Vector

magnitude
direction

- To display a Vector
- Vector Calculation
- Vector Application

Example



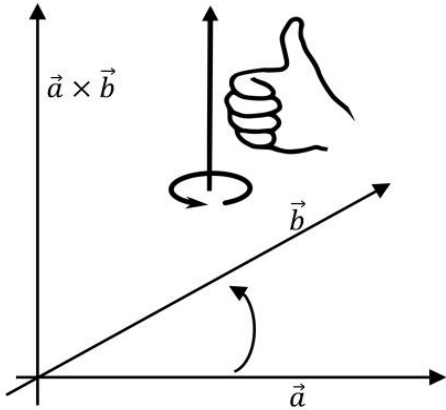
How to calculate the work?

$$W = F \cdot s \cos \theta$$

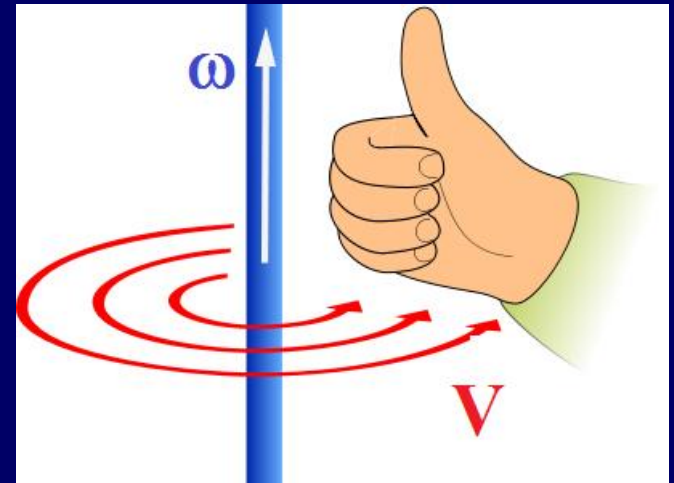
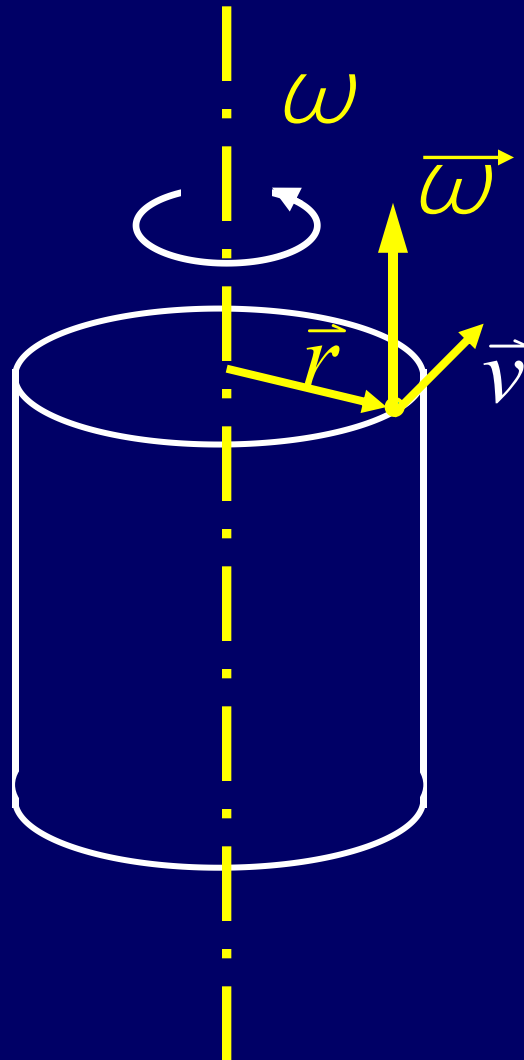
$$W = \vec{F} \cdot \vec{s}$$

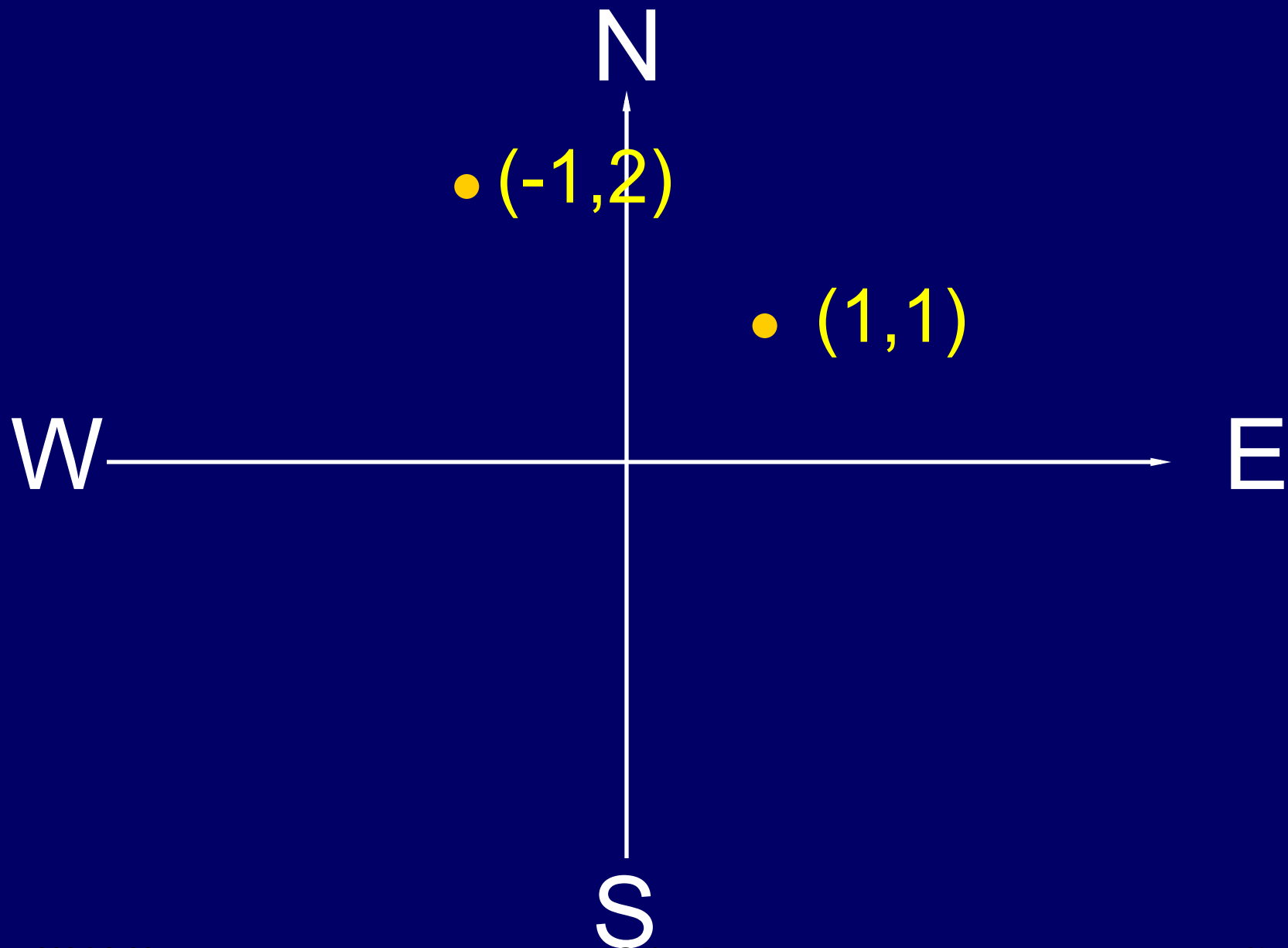
Example

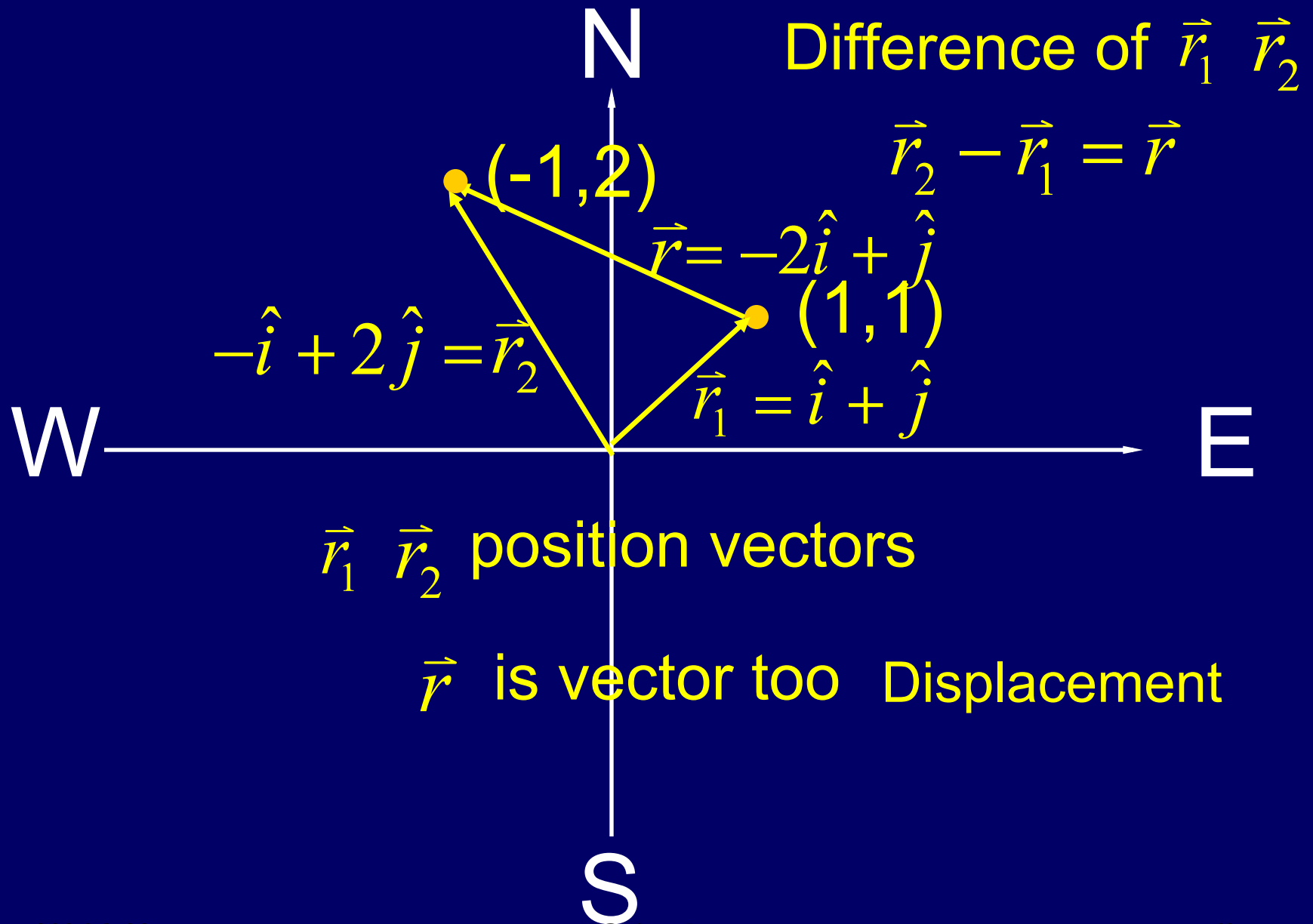
How to display the rotation?



$$\vec{v} = \vec{\omega} \times \vec{r}$$







Position, velocity and Acceleration

$\vec{r}$ is function of time t : $\vec{r} = \vec{r}(t)$.

$$\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$$

$$\vec{r}'(t') = x(t')\hat{i} + y(t')\hat{j} + z(t')\hat{k}$$

$$\Delta\vec{r} = \vec{r}'(t) - \vec{r}(t)$$

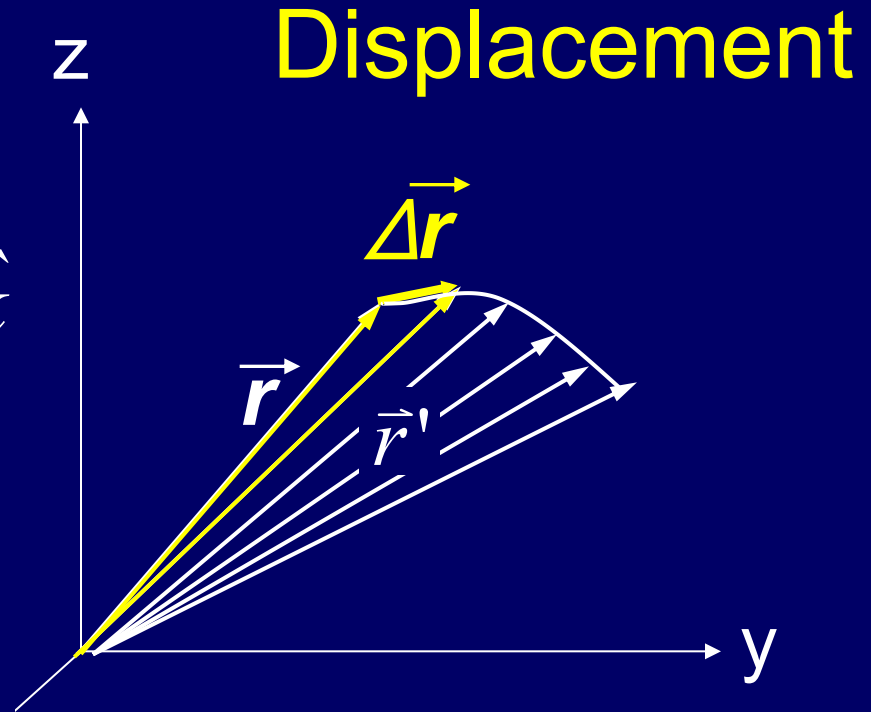
$$= \Delta x\hat{i} + \Delta y\hat{j} + \Delta z\hat{k}$$

$$\frac{\Delta\vec{r}}{\Delta t} = \bar{\vec{v}} \quad \text{Average Velocity}$$

$\Delta t \rightarrow 0$

Instantaneous Velocity

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta\vec{r}}{\Delta t} = \vec{v} = \frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k}$$



Position, velocity and Acceleration

$$\vec{v}(t) = v_x(t)\hat{i} + v_y(t)\hat{j} + v_z(t)\hat{k}$$

$$\vec{v}'(t') = v_x(t')\hat{i} + v_y(t')\hat{j} + v_z(t')\hat{k}$$

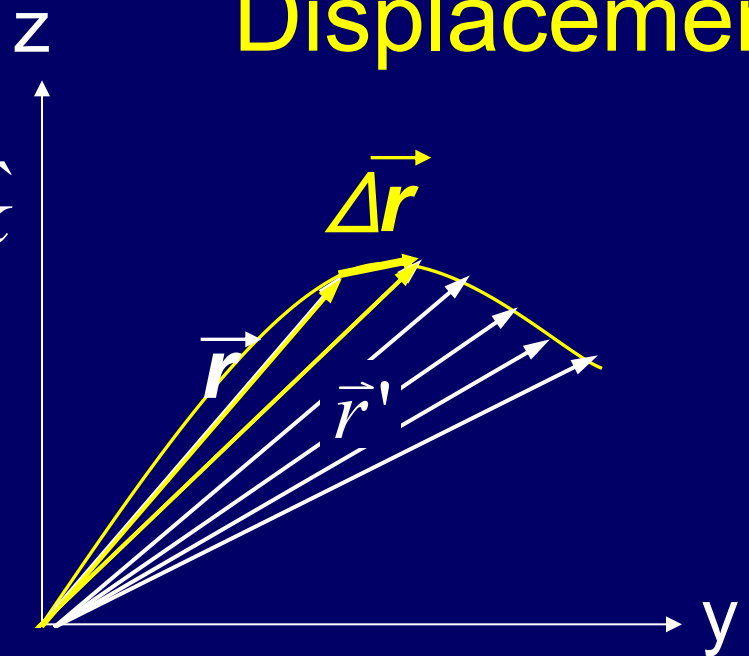
$$\Delta\vec{v} = \vec{v}'(t) - \vec{v}(t)$$

$$= \Delta v_x \hat{i} + \Delta v_y \hat{j} + \Delta v_z \hat{k}$$

$$\frac{\Delta\vec{v}}{\Delta t} = \bar{\vec{a}} \quad \text{Average acceleration}$$

$$\text{Instantaneous acceleration} \quad \lim_{\Delta t \rightarrow 0} \frac{\Delta\vec{v}}{\Delta t} = \vec{a} = \frac{d^2\vec{r}}{dt^2} = \frac{d^2x}{dt^2}\hat{i} + \frac{d^2y}{dt^2}\hat{j} + \frac{d^2z}{dt^2}\hat{k}$$

Displacement



Motion in one dimension



$x = x(t)$ $v(t) = \frac{dx(t)}{dt}$ $a(t) = \frac{dv(t)}{dt} = \frac{d^2x(t)}{dt^2}$

$$v = \frac{dx}{dt} = B + 2Ct$$

$$a = \frac{dv}{dt} = 2C$$

$$X = A + Bt + Ct^2$$

$$v = \frac{dx}{dt} = B$$

$$X = A + Bt$$

$$a = \frac{dv}{dt} = 0$$

$$v = \frac{dx}{dt} = 0$$

$$X = A$$

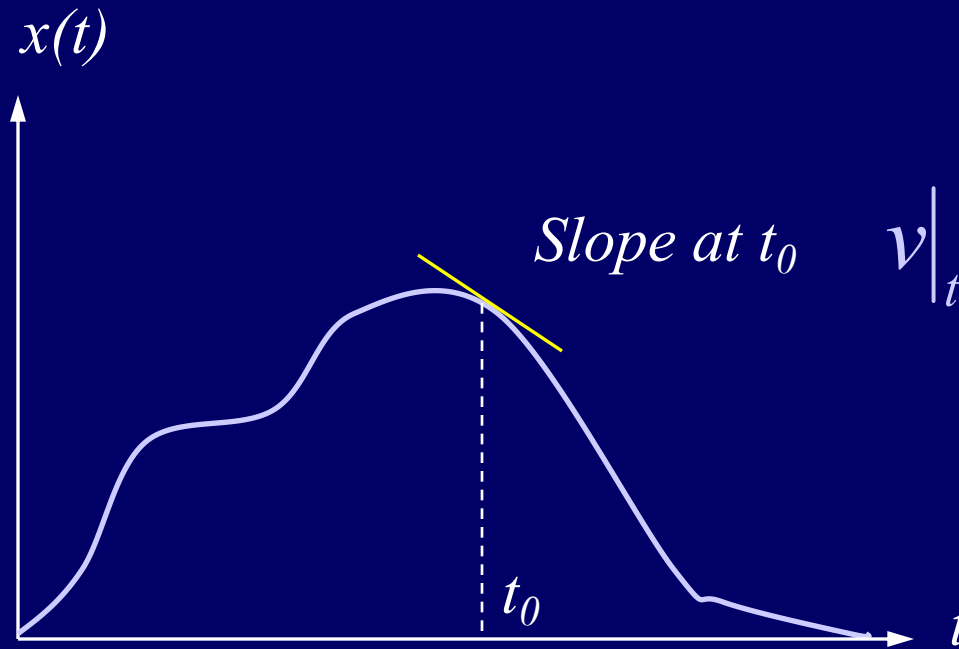
$$a = \frac{dv}{dt} = 0$$

A

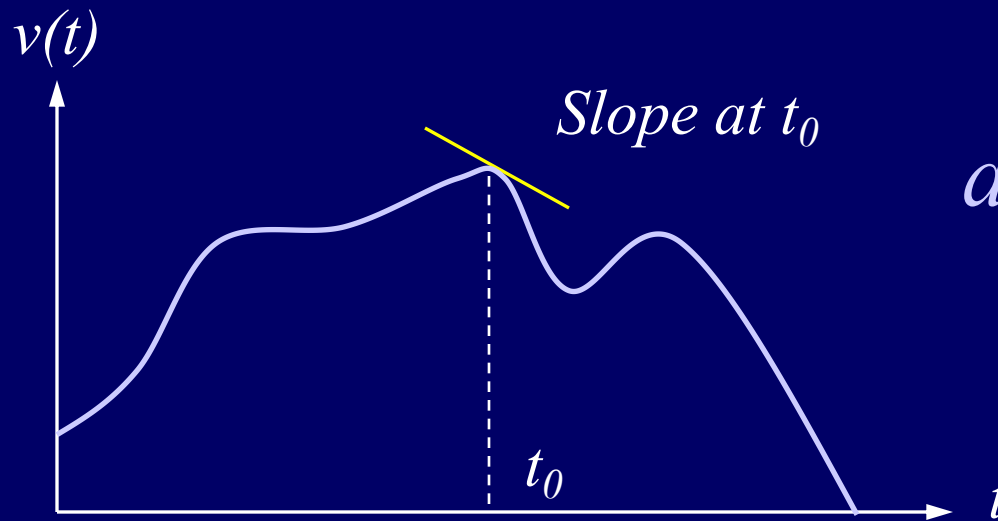
$$X = D \sin(\omega t + \phi)$$

t

$$a = \frac{dv}{dt} = -D\omega^2 \sin(\omega t + \phi) \quad v = \frac{dx}{dt} = D\omega \cos(\omega t + \phi)$$



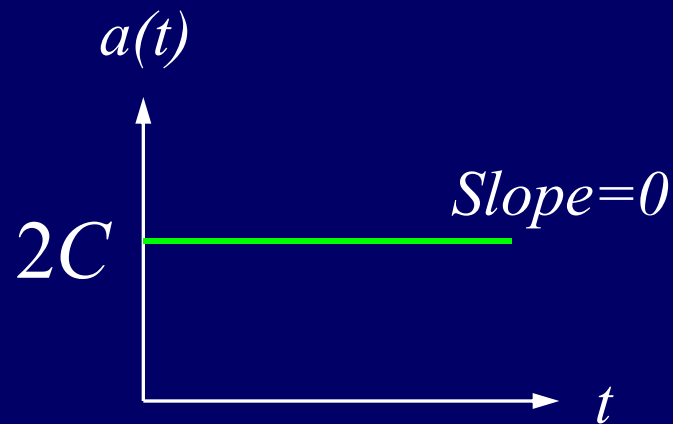
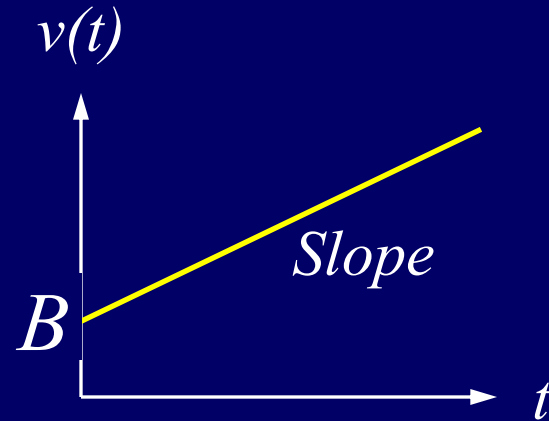
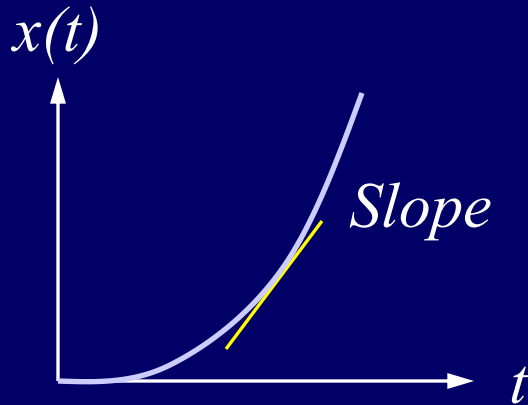
$$v|_{t=t_0} = \left. \frac{dx}{dt} \right|_{t=t_0}$$



$$a_{t=t_0} = \left. \frac{dv}{dt} \right|_{t=t_0}$$

Motion with constant acceleration

$$x=A+Bt+Ct^2 \quad v=\frac{dx}{dt}=B+2Ct \quad a=\frac{dv}{dt}=2C$$



$$\frac{da}{dt}=0$$

Motion with constant acceleration

$$\vec{a} = \frac{d\vec{v}}{dt} \quad d\vec{v} = \vec{a}dt \quad \int_{v_0}^v d\vec{v} = \int_{t_0}^t \vec{a}dt$$

$$v - v_0 = a(t - t_0) \quad v = v_0 + at \quad (t_0 = 0)$$

$$\vec{v} = \frac{d\vec{r}}{dt} \quad d\vec{r} = \vec{v}dt \quad \int_{r_0}^r d\vec{r} = \int_0^t \vec{v}dt$$

$$\int_{r_0}^r d\vec{r} = \int_0^t (v_0 + at)dt \quad r - r_0 = v_0t + \frac{1}{2}at^2$$

$$2a(r - r_0) = v^2 - v_0^2$$

Freely Falling Bodies



$$a = -g \quad g = 9.8m/s^2$$

$$\frac{dv}{dt} = -g \quad \int_0^v dv = \int_0^t -g dt \quad v = -gt$$

$$v = \frac{dy}{dt} \quad \int_0^y dy = \int_0^t v dt = \int_0^t -g t dt = -\frac{1}{2} g t^2$$

$$y = -\frac{1}{2} g t^2$$

Chapter 1. Exercises:29,31
Problems:7, 9

Chapter 2 Exercises:6,9,15
Problems:3, 18

Additional Problems

1. A point moves uniformly on a plane curve trajectory with velocity v . The magnitude of acceleration on a certain point of the trajectory is a . What is the curvature radius in that point?
2. A cyclist travels at 10 km/h heading north. Wind blows with a speed (relative to ground) of 6 km/h from a direction between N and E. To the cyclist the wind appears to come from the direction at 15° from North to East. (a) Find the speed of the wind relative to the cyclist and the direction of the wind, relative to ground. When the cyclist goes back, which are velocity and apparent direction of the wind (wind did not vary)?
3. An airplane is flying at constant velocity V , of horizontal direction and magnitude $V = 100$ m/s at the height $h = 5000$ m. A (super)cannon on earth shoots against it a ball at the moment in which the plane is just above the weapon. The velocity of the ball is $v_0 = 500$ m/s. Neglecting the presence of air, find: (a) the angle α at which we must shoot to hit the plane; (b) the time of the collision (which of the two solution should be chosen?); (c) how much did the plane travel up to this moment.
4. Two spheres, one with mass double that of the other, are launched upwards with the same initial velocity v_0 . If the resistance of air can be neglected, what is the ratio of the heights they reach? What is the ratio if there are the same initial momentum p_0 ?